The empirical recurrence for OEIS A297886, recovered from its tail

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Abstract

OEIS A297886 records "Empirical recurrence of order 86" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 138$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 92$. Its last coefficient is $c_{86} = 64 \neq 0$, so no further tail satisfies anything shorter; any order-86 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A297886 is "Number of nX5 0..1 arrays with every element equal to 0, 1, 2 or 4 king-move adjacent elements, with upper left element zero.". It has offset 1 and begins

$$16, 69, 41, 84, 114, 162, 248, 368, \ \ldots$$

The entry states:

> Empirical recurrence of order 86 (see link above)

As of the "Last modified" line on the live entry (Jan 07 2018 16:30:29, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 138$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 138$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 86: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 86.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 276$ exact terms from $n = 6$ on, it returns order 86 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{86} c_i\, a(n-i),$$

c_1	1	c_{23}	−174	c_{45}	−1577	c_{67}	−2674
c_2	0	c_{24}	310	c_{46}	2761	c_{68}	626
c_3	7	c_{25}	11	c_{47}	1098	c_{69}	3776
c_4	−3	c_{26}	356	c_{48}	2432	c_{70}	2598
c_5	−1	c_{27}	−270	c_{49}	−2965	c_{71}	−982
c_6	−18	c_{28}	279	c_{50}	−1422	c_{72}	−2022
c_7	−2	c_{29}	−437	c_{51}	−1976	c_{73}	−620
c_8	−7	c_{30}	387	c_{52}	3451	c_{74}	868
c_9	18	c_{31}	−979	c_{53}	3450	c_{75}	608
c_{10}	7	c_{32}	700	c_{54}	1389	c_{76}	−1148
c_{11}	39	c_{33}	−534	c_{55}	−2295	c_{77}	−1160
c_{12}	27	c_{34}	1227	c_{56}	−3164	c_{78}	−20
c_{13}	24	c_{35}	−341	c_{57}	−578	c_{79}	1120
c_{14}	−46	c_{36}	179	c_{58}	1410	c_{80}	1008
c_{15}	−94	c_{37}	−711	c_{59}	470	c_{81}	48
c_{16}	−132	c_{38}	46	c_{60}	−1078	c_{82}	−488
c_{17}	38	c_{39}	−384	c_{61}	−2319	c_{83}	−392
c_{18}	66	c_{40}	1298	c_{62}	216	c_{84}	−64
c_{19}	282	c_{41}	−137	c_{63}	2776	c_{85}	96
c_{20}	−34	c_{42}	852	c_{64}	2567	c_{86}	64
c_{21}	−111	c_{43}	−2563	c_{65}	118		
c_{22}	−233	c_{44}	−800	c_{66}	−4148		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 92$, and it is the recurrence OEIS A297886 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{86} - c_1 x^{86-1} - \cdots - c_{86}$. Its constant term is $-c_{86} = -64$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 86 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 86, so $\deg q = 86$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 86, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 34 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 86, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 64.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-86 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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